DATA SCIENCE TUTORIAL · PYTHON SERIES

Pandas for Data Analysis: 4 Skills for Every Project

TUTORIAL
#4

The tools that turn raw, messy data into actual answers —
without fighting the library.

@mistakenlyhuman

CONTEXT

The Problem With Most Pandas Tutorials

- They teach syntax, not mental models
- Real datasets aren’t clean CSVs
- You need tools for mess, not textbook examples

FOUNDATION

The Mental Model That Changes Everything

- A DataFrame is a structured, labeled, query-able object
- Columns are NumPy arrays with names
- Operations feel inevitable once you understand this

SKILL 1 · EXPLORATION

Explore Any Dataset in 60 Seconds

- `df.shape`, `df.dtypes`, `df.head()`
- `df.describe()` reveals outliers and skew instantly
- `df.isnull().sum()` — never skip this step

1

SKILL 2 · FILTERING

Select and Filter Without the Common Traps

- Use `&` and `|` not `'and'` /'or' — wrap each condition
- Switch to `.query()` at 3+ conditions
- Use `.isin()` instead of stacked OR conditions

2

SKILL 3 · CLEANING

Clean Messy Data Without Losing Rows

- `errors='coerce'` turns bad values into NaN, not errors
- Strip strings: `.str.strip().str.title()`
- Re-run `dtypes` and `isnull()` after every cleaning step

3

GroupBy and Aggregation

- Split → Apply → Combine is the mental model
- Use `.agg()` with named aggregations for clean output
- Always `reset_index()` to return to a flat DataFrame

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COMPARISON

GroupBy vs Pivot Table

GROUPBY		PIVOT TABLE
For computation and downstream processing	VS	For communication with stakeholders

> Same underlying data, different output shape

NEXT STEPS

What to Do Next

- Open a dataset you've been avoiding
- Run through all four skills in sequence
- Tutorial 5: Visualization — making the data visible

> *"The friction between raw data and real insight is where most data work actually happens — Pandas is the tool that makes that friction manageable instead of defeating."*